SAMITHA MUTHYALA

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PROFESSIONAL SUMMARY

Motivated BCA graduate with hands-on experience in Artificial Intelligence, Generative AI, Computer Vision, and secure software development. Experienced in building AI-powered applications using Python, LangChain, LangGraph, RAG, LLM APIs, video processing, pose estimation, and full-stack technologies. Currently developing an AI-based sports injury risk detection system through the Infosys Springboard Virtual AI Internship. Passionate about AI/ML, AI Agents, Computer Vision, Cybersecurity, and scalable software solutions.

EDUCATION

Christ Academy Institute for Advanced Studies — Bachelor of Computer Applications (BCA) – CGPA: 8.2/10 (82%) | Bengaluru | 2023 – 2026

Andhra Pradesh Board — Class XII – 77.74% | India | 2022

Andhra Pradesh Board — Class X – 100% | India | 2020

TECHNICAL SKILLS

Languages: Python, SQL, HTML, CSS, Java (Basics)

AI / ML: Artificial Intelligence, Machine Learning, Computer Vision, Pose Estimation, Video Processing

Generative AI: LangChain, LangGraph, RAG, Prompt Engineering, OpenAI API, Conversational AI, Multi-Agent Systems

Computer Vision: OpenCV, MediaPipe, Video Frame Analysis

Frameworks: Flask, Streamlit, React, Node.js, Express

Databases: MySQL, PostgreSQL, SQLite, FAISS

Tools: Git, GitHub, Linux, Jupyter Notebook, Render, Vercel, Wireshark

Core Concepts: Vector Databases, REST APIs, OOP, Data Structures, Secure Software Development

PROJECTS

AI Sports Injury Risk Detection | Python, Computer Vision, MediaPipe, OpenCV, React, Node.js | 2026

- Developing an AI-powered system that analyzes sports movement videos to identify movement patterns associated with injury risk.
- Built a video-upload workflow with backend processing, frame extraction, and pose-based movement analysis.
- Implemented a full-stack frontend/backend workflow for uploading videos and presenting injury-risk analysis results.

Customer Support RAG Agent with Memory | Python, LangChain, FAISS, Streamlit | 2026

- Built an AI-powered customer support chatbot using Retrieval-Augmented Generation (RAG).
- Created a FAISS vector database for semantic document retrieval and implemented conversational memory.
- Displayed source citations for AI-generated responses.

Multi-Agent Scheduling Assistant | Python, LangGraph, SQLite, OpenAI API | 2026

- Developed a LangGraph workflow with Triage and Booking agents.
- Built SQLite-backed scheduling tools with persistent memory and intelligent routing.
- Automated date normalization and webhook-based booking notifications.

SecureStego | React, TypeScript, Node.js, PostgreSQL | 2026

- Built a secure communication platform using cryptography and image steganography.
- Implemented AES-256 and RSA encryption before embedding messages into images.
- Developed secure encoding/decoding workflows and a responsive full-stack web application.

EXPERIENCE

Virtual AI Intern – Infosys Springboard | 2026 – Present | Virtual

Individual AI Project: Sports Injury Risk Detection Through Video

- Working on an individual AI project focused on detecting sports injury risk from movement videos.
- Applying computer vision, video processing, pose estimation, and AI/ML concepts to analyze human movement.
- Developing and integrating the frontend, backend, video-upload workflow, and analysis pipeline.
- Gaining practical experience in building, testing, debugging, and presenting an end-to-end AI application.

ACHIEVEMENTS

- Vice President – Centre for Indian Knowledge Systems, Christ Academy Institute for Advanced Studies (2025 – 2026). Organized seminars, workshops, and awareness programs; managed communication, scheduling, and documentation; developed leadership, teamwork, and public speaking skills.
- 2nd Prize – Internal Smart India Hackathon (2025)
- Presented a research project at a National-Level Conference (2025)
- Organized inter-college seminars and workshops under IKS

CERTIFICATIONS

- Google for Startups – Startup School: Prompt to Prototype
- PwC Cybersecurity Job Simulation
- Deloitte Cybersecurity Job Simulation
- Cyber Security Virtual Internship
- IoT / OT Hacking & Defensive Security

LANGUAGES

English (Fluent), Telugu (Native), Hindi (Fluent), Kannada (Conversational)